Z-residual Diagnostic Tool for Assessing Covariate Functional Form in Proportional Hazards Models with Shared Frailty

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Outline

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Section 1

Introduction

Shared Frailty Models

- The clustered survival data are often observed in survival analysis. A shared frailty model uses a common random effect to account for the correlation of clustered failure time data.

- Suppose there are g groups of individuals with n_i individuals in the ith group, $i = 1, 2, \ldots, g$.
  - T^*_{ij} is the true failure time of the ith cluster and the jth case; let t^*_{ij} be the realization of T^*_{ij}.
  - C_{ij} is the corresponding censoring time, assumed to be independent of T^*_{ij}.
  - $T_{ij} = \min(T^*_{ij}, C_{ij})$ is the observed failure time, which is possibly right-censored. Let t_{ij} be the realization of T_{ij}.
  - $\delta_{ij} = I(T^*_{ij} < C_{ij})$ is the indicator of non-censored T_{ij}.

Shared Frailty Models (contd.)

- For the proportional hazards model, the hazard function of T_{ij}^* is assumed to be

$$h_{ij}(t) = z_i \exp(\beta' x_{ij}) h_0(t) \tag{1}$$

  where x_{ij} is a vector of values of p explanatory variables for the jth individual in the ith group, β is the vector of their coefficients, $h_0(t)$ is the baseline hazard function, and the z_i is the frailty effect of the ith cluster.

- The survival function of T_{ij}^* is then given by

$$S_{ij}(t) = \exp\left\{ -\int_0^t h_{ij}(s)\,\mathrm{d}s \right\} = \exp\left\{ -z_i \exp(\beta' x_{ij}) H_0(t) \right\}, \tag{2}$$

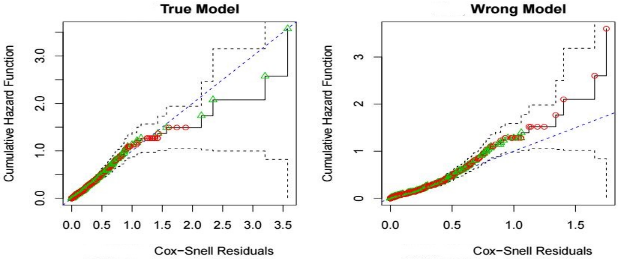
- We want to emphasize that the subscript ij of $S_{ij}(t)$ indicates that the survival function considers the covariate x_{ij} and cluster effect z_i.

Review of Existing Residuals: I

- Cox-Snell (CS) residual [1, 2]:

$$r^c_{ij}(t_{ij}) = -\log(\hat{S}_{ij}(t_{ij})), \tag{3}$$

where t_{ij} is the observed failure time. Then the Kaplan-Meier cumulative hazard function of $r^c_{ij}(t_{ij})$ is plotted to check whether they are exponentially distributed:



- One drawback to the Cox-Snell residuals is that they don't provide much insight into why the model's assumptions are violated.

Review of Existing Residuals: II

- Martingale residuals [3] are defined as:

$$r_{ij}^M = \delta_{ij} - r_{ij}^c, \tag{4}$$

  where δ_{ij} is the event indicator,

- Deviance residuals [4, 5] are defined as:

$$r_{ij}^D = sgn(r_{ij}^M)[-2(r_{ij}^M + \delta_{ij} \log(\delta_{ij} - r_{ij}^M))]^{\frac{1}{2}}, \tag{5}$$

  where r_{ij}^M is the martingale residual, the function $sgn(.)$ is the sign function.

- Martingale and deviance residual are most widely used for examining the validity of the function form of covariates.
- However, due to the censoring, there is no reference distribution for the above residuals. Therefore, it is challenging to develop numerical measures for quantifying the adequacy of the models with these residuals. The visual inspection of these residuals is often subjective.

What is Z-residual?

- The key idea is to replace the survival probability of a censored failure time with a uniform random number between 0 and the survival probability of the censored time, which results in randomized survival probability (RSP). It can be proved that RSPs are uniformly distributed on $(0, 1)$ [6]. A transformation of RSP with the quantile of standard normal will produce Z-residual, which is distributed as the standard normal under the true model.

- In this paper, we extend the concept of Z-residual and develop residual diagnostic tools tailored specifically for assessing the functional form of covariates in shared frailty models.

Section 2

Z-residual for Shared Frailty Models

Definition of Z-residual

- The randomized survival probability (**RSP**) for t_{ij} is defined as:

$$S_{ij}^{R}(t_{ij}, \delta_{ij}, U_{ij}) = \begin{cases} \hat{S}_{ij}(t_{ij}), & \text{if } t_{ij} \text{ is uncensored, } \delta_{ij} = 1, \\ U_{ij}\,\hat{S}_{ij}(t_{ij}), & \text{if } t_{ij} \text{ is censored, } \delta_{ij} = 0, \end{cases} \tag{6}$$

where U_{ij} is a uniform random number on $(0, 1)$, and $S_{ij}(\cdot)$ is the postulated survival function for T_{ij}^{*}. $S_{ij}^{R}(t_{ij}, \delta_{ij}, U_{ij})$ is a random number between 0 and $S_{ij}(t_{ij})$ when t_{ij} is censored.

- It is preferred to transfer **RSP** with the normal quantile:

$$r_{ij}^{Z}(t_{ij}, \delta_{ij}, U_{ij}) = -\Phi^{-1}(S_{ij}^{R}(t_{ij}, \delta_{ij}, U_{ij})). \tag{7}$$

where $\Phi()$ is the cumulative distribution function (CDF) of a standard normal distribution. We name the residuals in (7) as Z-residuals.

Independence and Uniformity of SP With No Censoring

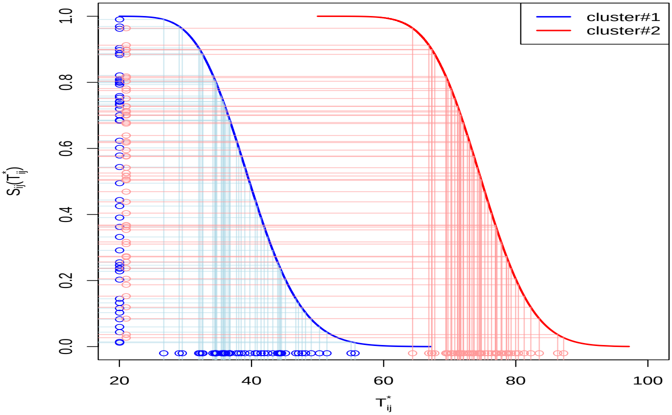
We first assume that there is no censoring and prove the uniformity and independence of $S_{ij}(T_{ij}^*)$ under the true model for shared frailty models. The survival probabilities, $S_{ij}(T_{ij}^*)$, as a function of the random variable T_{ij}^*, maintain uniformity over the interval $(0, 1)$. This uniformity can be proved as follows:

$$P(S_{ij}(T_{ij}^*) < t) = P(T_{ij}^* > S_{ij}^{-1}(t)) = S_{ij}(S_{ij}^{-1}(t)) = t, \tag{8}$$

where $t \in (0, 1)$. Conditional on the cluster indicator, $S_{ij}(T_{ij}^*)$, for $j = 1, \ldots, n_i$, are independent since T_{ij}^*, for $j = 1, \ldots, n_i$, are independent within a cluster and the T_{ij}^*'s across clusters are independent of each other.

A Graphical Illustration

Figure 1: Illustration of the uniformity of SPs based on synthetic data simulated from two gamma distributions for two clusters. The colours of the points depict their cluster indicators.



An Example

- Suppose T_{ij}^* follows a lognormal distribution and $\log(T_{ij}^*)$ is normally distributed with mean μ_{ij} and standard deviation σ_{ij}.
- In this scenario, the survival function $S_{ij}(T_{ij}^*)$ is given as

$$S_{ij}(T_{ij}^*) = 1 - \Phi\left(\frac{\log(T_{ij}^*) - \mu_{ij}}{\sigma_{ij}}\right)$$

- The Z-residual for this scenario can be derived as:

$$\begin{aligned} Z_{ij} &= -\Phi^{-1}(S_{ij}(T_{ij}^*)) = -\left[\Phi^{-1}\left(1 - \Phi\left(\frac{\log(T_{ij}^*) - \mu_{ij}}{\sigma_{ij}}\right)\right)\right] \\ &= \frac{\log(T_{ij}^*) - \mu_{ij}}{\sigma_{ij}} \end{aligned}$$

- This example shows the relationship between Z-residuals Z_{ij} and conditional Pearson Residuals.

Independence and Uniformity of RSPs With Censoring

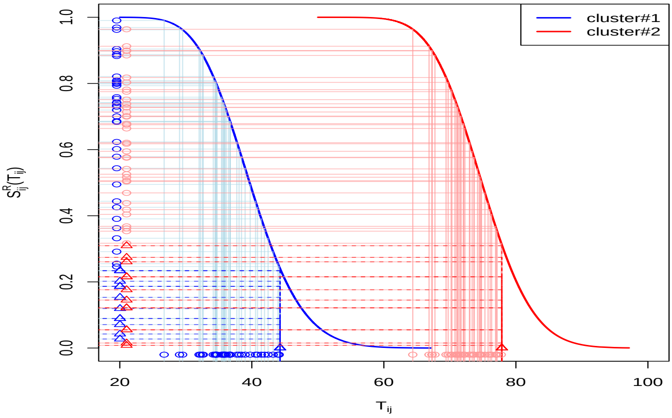
- We will assume that T^*_{ij} and C_{ij} are independent, indicating that C_{ij} is non-informative for the original failure times.
- We rewrite the RSP defined in equation (6) as a function of the following random variables: the original uncensored failure time T^*_{ij}, censoring time C_{ij}, and a uniform random number U_{ij}, as follows:

$$S^{R'}_{ij}(T^*_{ij}, C_{ij}, U_{ij}) = \begin{cases} S_{ij}(T^*_{ij}), & \text{if } T^*_{ij} \le C_{ij} \\ U_{ij}\, S_{ij}(C_{ij}), & \text{if } T^*_{ij} > C_{ij}. \end{cases} \tag{9}$$

- Given covariates and cluster indicators, T^*_{ij}'s are independently distributed and are independent of the censor the distribution of $S_{ij}(T^*_{ij})$ given $C_{ij} = c$ is also a uniform distribution. We therefore first fix C_{ij} to a value c
- More specifically, given $T^*_{ij} \le c$, the RSP, $S_{ij}(T^*_{ij})$ (equation (9)), is uniformly distributed on $(S_{ij}(c), 1)$.
- Given $T^*_{ij} > c$, the RSP is $U_{ij}\, S_{ij}(c)$, uniformly distributed on $(0, S_{ij}(c))$ due to the uniformity of U_{ij}.

A Graphical Illustration

Figure 2: Illustration of the uniformity of RSPs based on synthetic data simulated from two gamma distributions for two clusters. The colours of the points depict their cluster indicators.



Diagnosis of the Functional Form of Covariates using Z-residuals: I

- The conditional distribution of Z-residual given x_i is approximately a standard normal and is homogeneous at varying levels of covariates when a model is correctly specified.

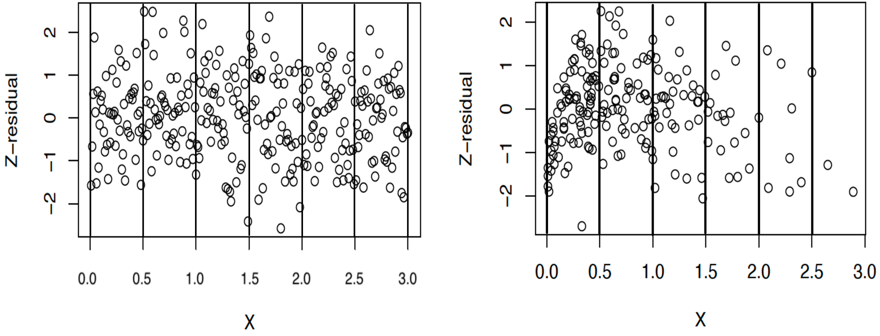
- To specifically check the functional form of covariates, we can plot Z-residuals against the covariates and linear predictors. In a correctly specified model, we expect no discernible trend in these scatterplots.

- The Z-residuals can be used to develop flexible tests for checking the overall goodness of fit (GOF) and identify specific model misspecification.

- A non-homogeneity test is proposed for testing whether there is a trend in Z-residuals for checking the covariate functional form.

Diagnosis of the Functional Form of Covariates using Z-residuals: II

Figure 3: An illustrative plot showing how to construct the non-homogeneity test with Z-residuals: dividing Z-residuals by a covariate or linear predictor (LP) with equally-spaced interval, then testing the equality of the means of grouped residuals. We apply the F-test in ANOVA to test the equality of the means of grouped Z-residuals.



A P-value Upper Bound for Assessing Replicated Z-residuals Test P-values

- Conducting statistical tests with Z-residuals can be challenging due to the randomness in the test p-values. To address this randomness, we can use an upper bound for the p-values. The process draws upon the distribution of order statistics of correlated random variables [7, 8], leading to an inequality denoted as $P(p_{(r)} < t) \leq \min\left(1, t\frac{J}{r}\right)$. This formulation enables the establishment of an upper bound for observed (simulated) rth statistics, denoted as $p_{(r)}$, which can be computed as $\min\left(1, p_{(r)}\frac{J}{r}\right)$. To avoid selecting a specific r, we calculate the minimal upper bound across $r = 1, \ldots, J$, denoted as $p_{\min}$:

$$p_{\min} = \min_{r=1,\ldots,J} \min\left(1, p_{(r)}\frac{J}{r}\right). \quad (10)$$

- $p_{\min}$ is advisable to use a threshold of 0.25 by Yuan and Johns [9], as opposed to the conventional 0.05 threshold for exact p-values, when declaring model inadequacy in practical contexts.

Censored Z-residuals and CZ-CSF test

- Survival probabilities can be transformed using the quantile of standard normal distribution [10], defined as

$$r_{ij}^{n}(t_{ij}) = -\Phi^{-1}(\hat{S}_{ij}(t_{ij})), \tag{11}$$

  $\hat{S}_{ij}(t_{ij})$ is estimated survival function of the jth individual from the ith cluster. We will call it **censored Z-residuals** in this study.
- The function `gofTestCensored` in R package `EnvStats` [11] provides an SF test that can be applied to check the normality of censored Z-residuals for checking the overall GOF of survival models. We will refer to this test using the **CZ-CSF** method in this study.
- CZ-CSF is only an overall GOF test, which gives non-random p-values. However, its power may not be as good as the tests targeted for testing the non-homogeneity of Z-residuals.

Section 3

Simulation Study

Data Generation and Simulation Setup

- True model:

$$h_{ij}(t) = z_i \exp(\beta_1 x_{ij}^{(1)} + \beta_2 \log(x_{ij}^{(2)}) + \beta_3 x_{ij}^{(3)}) h_0(t), \tag{12}$$

- Wrong model:

$$h_{ij}(t) = z_i \exp(\beta_1 x_{ij}^{(1)} + \beta_2 x_{ij}^{(2)} + \beta_3 x_{ij}^{(3)}) h_0(t), \tag{13}$$

- The true survival times $t_{ij}^* \sim$ Weibull baseline hazard function with shape (α) parameter is 3 and scale (λ) parameter is 0.007.

$$t_{ij}^* = \left\{ \frac{-\log(u_{ij})}{\lambda z_i \exp(x_{ij}^{(1)} - 2\log(x_{ij}^{(2)}) + 0.5 x_{ij}^{(3)})} \right\}^{(1/\alpha)} \tag{14}$$

- The censored time $C_{ij} \sim \text{Exp}(\theta)$, resulting in different censoring rates: 0%, 20%, 50% and 80%.
- g=20 and $n_i (n_i = J)$ from 10 to 100;
- For each combination of cluster size and censoring rate, we generated 1000 datasets for estimating model rejection rates of different diagnostics methods.

The Graphical Check

The graphical checks based on Z-residuals detect the overall GOF and adequacy of the functional form of covariates.

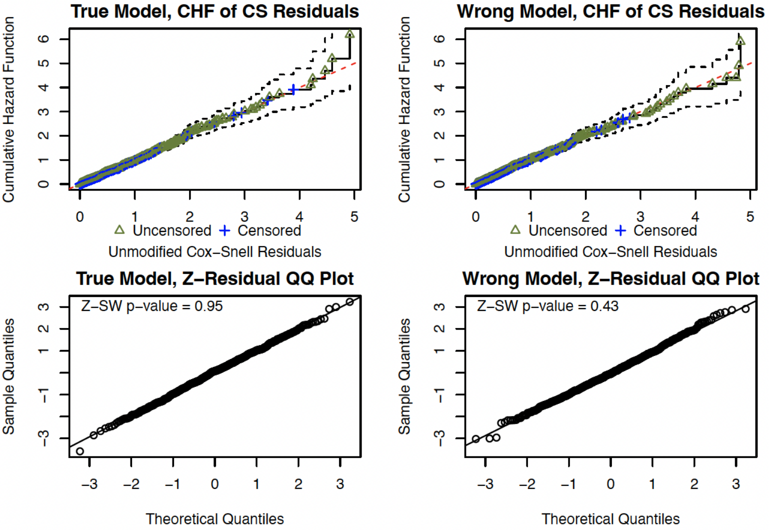
- The graphical methods used for assessing the overall GOF of the model included CHFs of CS residuals and quantile-quantile (QQ) plots of Z-residuals.

- The advantage of examining the scatterplots of Z-residuals against the linear predictor and covariates is that they are used to diagnose the misspecification of the functional form of covariates.

- The Z-residuals are divided into 10 groups by cutting the linear predictors and covariates into equally spaced intervals. The boxplot indicates that the Z-residuals are homogeneous across groups under the true model, but exhibit differential group means under the wrong model.

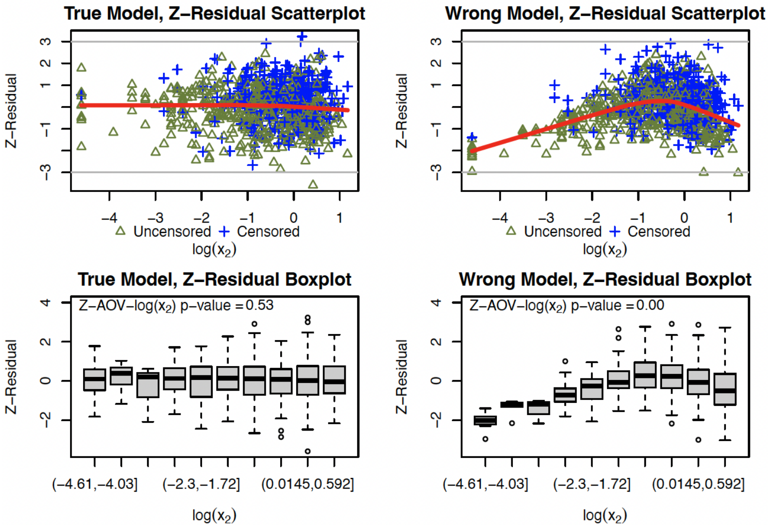
The overall GOF Graphical Check

Figure 4: Performance of Z-residuals as a graphical tool for assessing the overall GOF of the model. The dataset has a sample size n = 800 and a censoring rate $c \approx 50\%$.



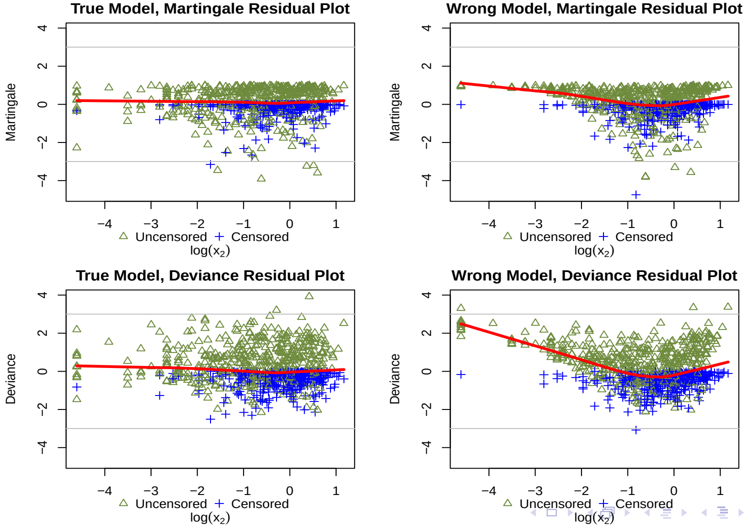
The Scatterplot and Boxplot of Z-residuals Residuals

Figure 5: Performance of Z-residuals as graphical tools for detecting the misspecification of the functional form of covariates. The dataset has a sample size n = 800 and a censoring rate $c \approx 50\%$.



The Scatterplot of Martingale and Deviance Residuals

Figure 6: Performance of the martingale and deviance residuals as a graphical tool for checking the functional form of covariates. The dataset has a sample size n = 800 and a censoring rate $c \approx 50\%$.



Numerical Tests

To complement the graphical assessment, numerical tests with Z-residuals can be used, as Z-residuals are approximately distributed as the standard normal under the true model.

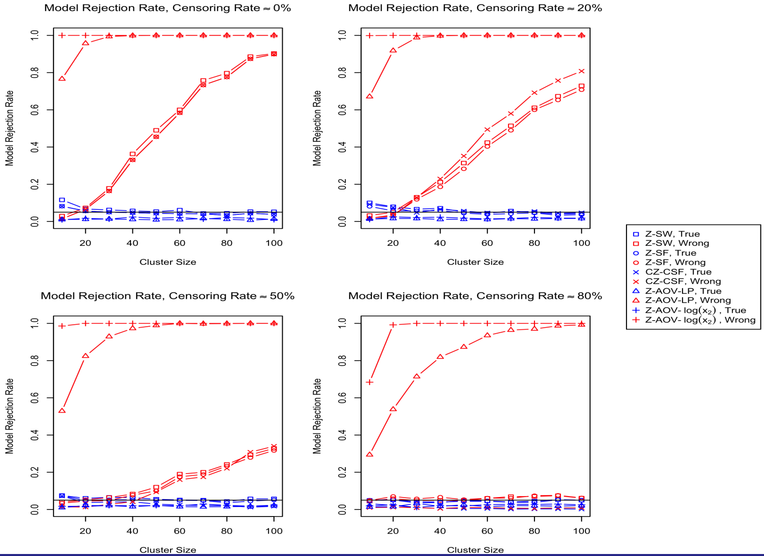
- Shapiro-Wilk (SW) or Shapiro-Francia (SF) normality tests can be applied to Z-residuals or censored Z-residuals to conduct numerical tests for the overall GOF. These tests are denoted by Z-SW, Z-SF, CZ-CSF.

- We can also divide Z-residuals into groups by cutting the linear predictor or a covariate into equally-spaced intervals and test the equality of means of Z-residuals across these groups using ANOVA. These tests are denoted by Z-AOV-LP or Z-AOV-X.

We generated 1000 datasets for each combination of cluster size and censoring rate. The Z-SW, Z-SF, CZ-CSF, Z-AOV-LP, and Z-AOV-X methods were assessed by estimating their model rejection rates with the proportions of test p-values less than 0.05.

Model Rejection Rates

Figure 7: Model rejection rates of various statistical tests based on Z-residual. A model is rejected when the test p-value is smaller than 5%.



Section 4

Real Data Demonstration

The Dataset

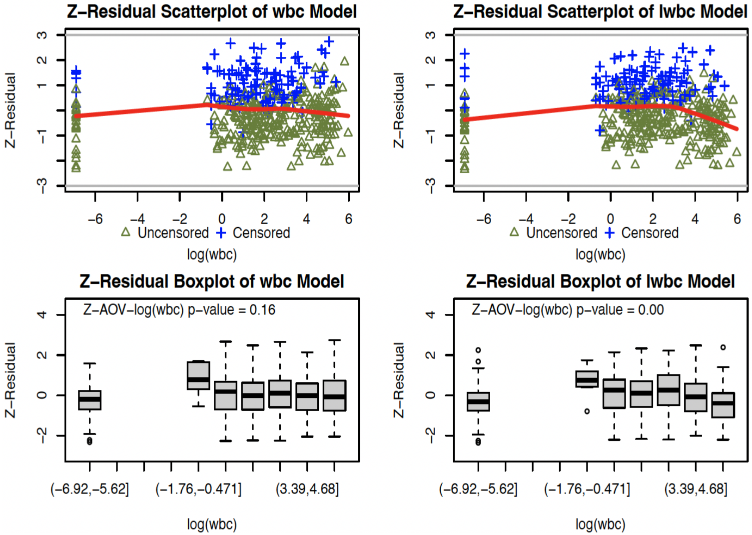
- A study [12] was conducted on 411 patients with acute myeloid leukemia to investigate the association between survival time and prognostic factors, including age, sex, white blood cell count (wbc) at diagnosis, and the townsend score (tpi). The study had a censoring rate of 29%.

- Two shared frailty models were fitted:

  1. wbc model: survival time $\sim$ wbc + age + sex + tpi
  2. lwbc model: survival time $\sim$ log(wbc) + age + sex + tpi

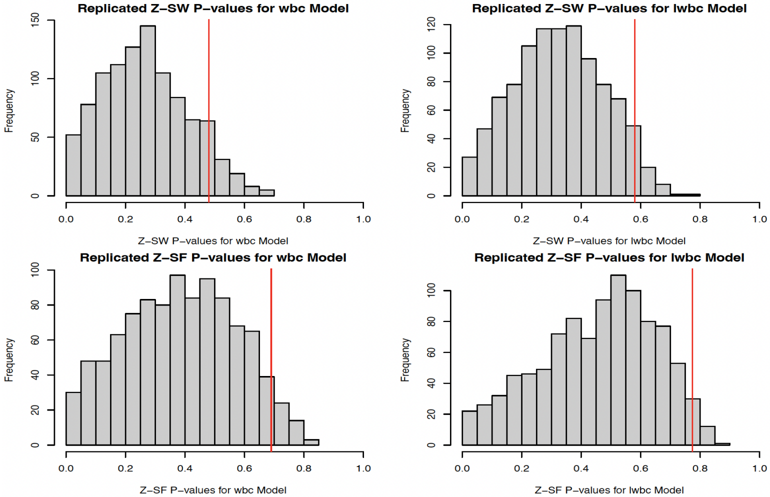
The Scatterplot and Boxplot of Z-residuals Residuals

Figure 8: Diagnostics results for the wbc (left panels) and lwbc (right panels) models fitted to the survival data of acute myeloid leukemia patients.



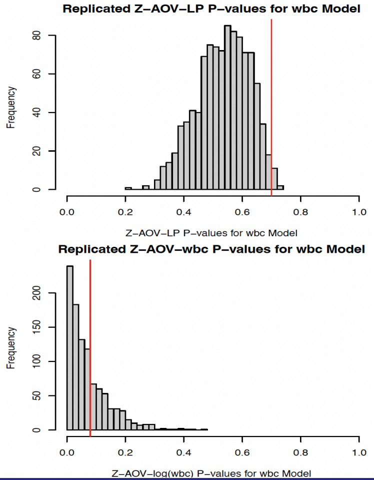
The Histograms of Replicated Z-SW and Z-SF P-Values

Figure 9: The histograms of 1000 replicated Z-SW and Z-SF p-values for the wbc model (left panels) and the lwbc model (right panels) fitted with the survival times of acute myeloid leukemia patients. The vertical red lines indicate p_{min} for 1000 replicated p-values.



The Histograms of Replicated Z-AOV-LP and Z-AOV-log(wbc) P-Values

Figure 10: The histograms of 1000 replicated Z-AOV-LP and Z-AOV-log(wbc) p-values for the wbc model (left panels) and the lwbc model (right panels). The vertical red lines indicate p_{min} for 1000 replicated p-values.



The Table of AIC, P-Values or $p_{\min}$ Values

Table 1: AIC, p-values or $p_{\min}$ values for the CZ-CSF test, $p_{\min}$ for Z-SW, Z-SF, Z-AOV-LP and Z-AOV-log(wbc) test for the wbc and lwbc models, respectively, for the acute myeloid leukemia data.

Model	AIC	CZ-CSF $p-$value	Z-SW $p_{\min}$	Z-SF $p_{\min}$	Z-AOV-LP $p_{\min}$	Z-AOV-log(wbc) $p_{\min}$
wbc model	3111.669	0.255	0.495	0.693	0.703	0.074
lwbc model	3132.105	0.305	0.579	0.781	0.978	$<$**0.00001**

Section 5

Package, Demonstration, and References to Z-Residual

Package, Demonstration, and References to Z-Residual

- The Z-residual package can be downloaded from https://github.com/tiw150/Zresidual.
- We have also provided a detailed demonstration of how to use this package for detecting the covariate functional form, available from this link: https://tiw150.github.io/Zresidual_demo.html.
- References to Z-Residual:

  - Wu, T., Li, L., Feng, C. (2023). Z-residual diagnostics for detecting misspecification of the functional form of covariates for shared frailty models. Journal of Applied Statistics, early view available online. Preprint available on arXiv: https://arxiv.org/abs/2302.09106
  - Li, L., Wu, T., Feng, C. (2021). Model Diagnostics for Censored Regression via Randomized Survival Probabilities. Statistics in Medicine, 40:2, 1482-1497. The accepted version available on arXiv: https://arxiv.org/abs/1911.00198

Section 6

Conclusions

Conclusions

- The proposed Z-residuals offer valuable insights into model adequacy and provide both graphical and numerical methods to examine the fit of the model to the data.
  1. By plotting Z-residuals against covariates, we can visually inspect the presence of trends, allowing us to assess the appropriateness of the functional form assumptions.
  2. The non-homogeneity tests based on grouped Z-residuals enables us to quantify the degree of mis-specification of covariate functional forms. Our simulation studies show that they have significantly higher power than the overal GOF tests based on Z-residual or censored Z-residuals.

- Z-residuals offer a powerful advantage in detecting specific model misspecifications that might be overlooked by traditional overall model diagnosis.

Future Works

- The double use of the data in both model parameter estimation and residual calculation may reduce the power of detecting model misspecification. Cross-validation could be a good method to solve this problem. By employing cross-validatory Z-residuals, we can achieve more robust and unbiased assessments of model adequacy, improving the accuracy and reliability of our diagnostic framework.

- We are working on extending Z-residual diagnostics to Bayesian models, which seems a more interesting topic.

Section 7

References

References

[1] D. R. Cox and E. J. Snell. A general definition of residuals. *Journal of the Royal Statistical Society. Series B, Methodological*, 30(2):248–275, 1968.

[2] David Collett. *Modelling Survival Data in Medical Research*. Chapman and Hall/CRC, 2015.

[3] Terry M. Therneau, Patricia M. Grambsch, and Thomas R. Fleming. Martingale-based residuals for survival models. *Biometrika*, 77(1):147–160, March 1990.

[4] Terry M. Therneau and Patricia M. Grambsch. *Modeling Survival Data: Extending the Cox Model*. Springer Science & Business Media, November 2013.

[5] P. McCullagh and John A. Nelder. *Generalized Linear Models, Second Edition*. CRC Press, August 1989.

References (contd.)

[6] Longhai Li, Tingxuan Wu, and Cindy Feng. Model diagnostics for censored regression via randomized survival probabilities. *Statistics in medicine*, 40(6):1482–1497, 2021.

[7] G Caraux and O Gascuel. Bounds on distribution functions of order statistics for dependent variates. *Statistics & probability letters*, 14(2):103–105, 1992.

[8] Tomasz Rychlik. Stochastically extremal distributions of order statistics for dependent samples. *Statistics & probability letters*, 13(5):337–341, 1992.

[9] Ying Yuan and Valen E. Johnson. Goodness-of-Fit Diagnostics for Bayesian Hierarchical Models. *Biometrics*, 68(1):156–164, 2012.

[10] Alessandra Nardi and Michael Schemper. New Residuals for Cox Regression and Their Application to Outlier Screening. *Biometrics*, 55(2):523–529, June 1999.

References (contd.)

[11] Steven P. Millard. *EnvStats: An R Package for Environmental Statistics*. Springer, New York, NY, 2nd ed. 2013.. edition, 2013.

[12] E. H ESTEY, YU SHEN, and P. F THALL. Effect of time to complete remission on subsequent survival and disease-free survival time in aml, raeb-t, and raeb. *Blood*, 95(1):72–77, 2000.

Thank you for your attention!
Questions and comments are welcome!